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United States Patent Application Publication

20250276840

Kind Code

A1

Publication Date

September 04, 2025

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Contaminant Resistant Product Packaging (Con)

Abstract

Embodiments are directed toward a product package filter assembly that prevents ambient contaminants from entering into the sealable product container. In some embodiments, the product package filter assembly includes an optional support element, a filter element, and a surface film element that are layered together with the filter element between the support element and the surface film element. The perimeters of the support element, the filter element, and the surface film element are sealably joined together. The product package filter assembly is then sealably joined over an opening of a product container. During heating of a product residing in the product container, gases generated in the product container vent out through the product package filter assembly, while the filter element prevents entry of contamination particles into the product container.

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Family ID: 95581204

Appl. No.: 18/887304

Filed: September 17, 2024

Related U.S. Application Data

parent US continuation-in-part 18385829 20231031 parent-grant-document US 12091231 child US 18887304

Publication Classification

Int. Cl.: B65D81/20 (20060101); B65D81/26 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This non-provisional application is a continuation-in-part of co-pending non-provisional application Ser. No. 18/385,829, filed Oct. 31, 2023, titled CONTAMINANT RESISTANT PRODUCT PACKAGING and claims the benefit of the filing date of such application under 35 U.S.C. § 120, the entirety of which is hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present disclosure relates generally to product packaging. Known product packaging is not satisfactory for situations where contamination from ambient contaminants is of concern. For example, a gluten free product cannot be processed in an ambient environment that is not entirely free of gluten particles. In such situations where cooking, heating, storage, and/or handling occurs in an ambient environment with gluten contamination, the gluten free product may become contaminated with gluten particles from the ambient environment during processing and/or during handling of the gluten free product.

[0003] As another example, some individuals are highly allergic to peanuts or other types of particles. When a meal is prepared for such at-risk individuals, special procedures and/or equipment is required for safe food preparation so that the prepared meal is not contaminated with any peanut particles or other particles of interest.

[0004] As yet another example, biological contaminants such as bacteria and/or viruses might enter into a product package. For example, food may be prepared and then transported to another location for consumption. Biological contaminants might enter into the product package during transportation, thereby contaminating the food and placing the consumer's health at risk.

[0005] As yet another example, some non-food product items may need to be sterilized in a package such that ambient contaminants do not enter the product package during and/or after sterilization. An example is sterilization of medical equipment.

[0006] Accordingly, there exists a need in the product packaging arts for improved product packaging that protects packaged products for contaminants.

SUMMARY OF THE INVENTION

[0007] The present disclosure is directed to embodiments of a product package filter assembly that prevents ambient contamination particles from entering into the sealable product container. In some embodiments, the product package filter assembly includes an optional support element, a filter element, and a surface film element that are layered together with the filter element between the support element and the surface film element. The perimeters of the support element, the filter element, and the surface film element are sealably joined together. The product package filter assembly is then sealably joined over an opening of a product container. During heating of a product residing in the product container, gasses generated in the product container vent out through the product package filter assembly, while the filter element prevents entry of any ambient contamination particles into the product container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Preferred and alternative embodiments of the present invention are described in detail below

with reference to the following drawings.

[0009] FIG. 1 is an exploded view of a product package filter assembly.

[0010] FIG. 2 is a perspective view of the product package filter assembly with the support element, the filter element and the surface film element of FIG. 1 sealably joined together.

[0011] FIG. 3 is a perspective view of the product package filter assembly and the product container after they have been sealably joined together.

[0012] FIG. 4 is an exploded view of a product package filter assembly that further comprises a removeable moisture barrier film.

[0013] FIGS. 5A-5B are a cross sectional views of the product container, the sealably joined support element, filter element, and surface film element, and the removeable moisture barrier film.

[0014] FIG. 6 is a perspective view of a product container that has a relatively small opening disposed on the surface of the product container.

[0015] FIG. 7 is a perspective view of a gusseted type product container provisioned with a product package filter assembly sealably joined to a port or opening disposed in the gusset bag.

[0016] FIG. 8 is a perspective view of a stand-up type product container provisioned with a product package filter assembly sealably joined to a port or opening disposed in the stand-up type bag.

[0017] FIG. 9A is a perspective view of a product package filter assembly wherein a length of the support element is greater than the length of the filter element and the surface film element.

[0018] FIG. 9B is a cross sectional view of the product package filter assembly of

[0019] FIG. 9A secured to a tray type product container.

[0020] FIG. 10A is a perspective view of a product package filter assembly wherein a width of one side of the support element is greater than the corresponding width of the filter element and the surface film element.

[0021] FIG. 10B is a perspective view of the product package filter assembly of FIG. 10A.

[0022] FIG. 11 is an exploded view of an alternative embodiment of the product

[0023] package filter assembly

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0024] Heating of a product residing in a product container generates steam and/or other gases (interchangeably referred to herein as a gas). Heating as generically referred to herein corresponds to any temperature change that may occur within a package and/or a product within the package. A temperature increase will expand any gasses residing in the package. A temperature increase in the product will expand any gases in the product and/or generate vapor with will expand with increasing temperature. Cooling will result in a volume decrease in any gas within the package (interchangeably referred to herein as a product container). In some situations, the product may absorb moisture during cooling. The temperature change (heating or cooling) may be intentionally induced, such as by heating in an oven, heating in a microwave, heating on a stove top element, heating in an autoclave, cooling in a refrigerator, or the like. In some situations, the temperature change may occur due to naturally occurring changes in ambient air temperature. All such induced temperature changes or naturally occurring temperature changes are contemplated herein.

[0025] When a contaminant filter assembly **100** embodiment is sealably joined to an opening in the product container, the contaminant filter assembly **100** permits expanding gas within the package and/or that is generated by the heating of the product to vent out from the product package filter assembly into an ambient region while preventing ambient contamination particles from entering into the interior of the product container. Conversely, cooling will decrease the volume of any gas within the package. Accordingly, the contaminant filter assembly **100** permits entry of ambient gas (air) into the package. Further, moisture may pass through the contaminant filter assembly **100** and enter into the package during cooling.

[0026] Embodiments of a disclosed product package filter assembly **100** will become better understood through review of the following detailed description in conjunction with the figures. The detailed description and figures merely provide examples of the various invention

embodiments described herein. Those skilled in the art will understand that the disclosed example embodiments may be varied, modified, and altered without departing from the scope of the invention as described herein. Many variations are contemplated for different applications and design considerations. However, for the sake of brevity, each and every contemplated variation is not individually described in the following detailed description.

[0027] Throughout the following detailed description, example embodiments of various product package filter assembly **100** embodiments are provided. Related features in the example embodiments may be identical, similar, or dissimilar in different examples. For the sake of brevity, related features will not be redundantly explained in each example. Instead, the use of related feature names will cue the reader that the feature with a related feature name may be similar to the related feature in an example explained previously. Features specific to a given example will be described in that particular example. The reader should understand that a given feature need not be the same or similar to the specific portrayal of a related feature in any given figure or example embodiment.

[0028] As used herein, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The terms “front,” “forward,” “rear,” and “rearward” are defined relative to an identified front element. The terms “height,” “vertical,” “upper,” “lower,” “above,” “below,” “top,” “bottom,” “topmost,” and “bottom-most” are defined relative to a vertical axis of a product package filter assembly **100**. The vertical axis is non-parallel to the longitudinal axis and is defined as parallel to the direction of the earth's gravity force on the product package filter assembly **100** when the product package filter assembly **100** is on horizontal ground. The term “lateral” is defined relative to the lateral axis of the product package filter assembly **100**. The lateral axis is non-parallel to the longitudinal and vertical axes of the product package filter assembly **100**. The longitudinal axis is referenced to a longest length, if present, of the product package filter assembly **100**.

[0029] The following definitions apply herein, unless otherwise indicated. “Substantially” means to be more-or-less conforming to the particular dimension, range, shape, concept, or other aspect modified by the term, such that a feature or component need not conform exactly. For example, a “substantially cylindrical” object means that the object resembles a cylinder, but may have one or more deviations from a true cylinder.

[0030] The term “configured” as used herein means an element being one or more of sized, dimensioned, positioned, or oriented to achieve or provide the recited function or result. The term “directly coupled” as used herein means that a component contacts (for example, when bolted) another component. The term “indirectly coupled” as used herein means that a first component is coupled to a second component by way of one or more intervening components that are directly coupled to the first and second components. A first component that is indirectly coupled to a second component is directly coupled to a third component, which may be directly coupled to the second component or to a fourth component that is directly coupled to the second component. “Coupled” means connected, either permanently or releasably, whether directly or indirectly, through intervening components. The term “coupled” should therefore be understood to disclose both direct and indirect coupling of components or elements that are described as being coupled to each other.

[0031] The term to “sealably join” or the like is defined to mean that two adjacent surfaces of one or more elements of the product package filter assembly **100**, and/or a product container, are affixed together (sealed together, or joined) in a manner such that fluids and/or gasses cannot pass between the two or more joined surfaces. That is, when two elements are sealably joined together, gas and/or liquids cannot pass through the sealing means and the two or more sealably joined surfaces. A variety of sealing means may be used in the various embodiments of the contaminant filter assembly **100**. An example sealing means includes a bead or strip of adhesive. Any suitable sealing means may be used in alternative embodiments. Such sealing means include, but are not limited to, crimping, heat sealing, induction sealing, pleating, sewing, or the like.

[0032] “Comprising,” “including,” and “having” (and conjugations thereof) are used interchangeably to mean including but not necessarily limited to, and are open-ended terms not intended to exclude additional elements or method steps not expressly recited. Terms such as “first”, “second”, and “third” are used to distinguish or identify various members of a group, or the like, and are not intended to denote a serial, chronological, or numerical limitation.

[0033] The term “or” is an inclusive grammatical conjunction to indicate that one or more of the connected terms may be employed. For example, the phrase “one or more A, B, or C” or the phrase “one or more As, Bs, or Cs” is employed to discretely disclose each of the following: i) one or more As, ii) one or more Bs, iii) one or more Cs, iv) one or more As and one or more Bs, v) one or more As and one or more Cs, vi) one or more Bs and one or more Cs, and vii) one or more As, one or more Bs, and one or more Cs. The term “based on” as used herein is not exclusive and allows for being based on additional factors not described. The articles “a,” “an,” and “the” include plural references. Plural references are intended to also disclose the singular.

[0034] FIG. 1 is an exploded view of a product package filter assembly **100** before fabrication. Embodiments of the product package filter assembly **100** comprise an optional support element **102**, a filter element **104**, and a surface film element **106**.

[0035] The support element **102** is defined by a support element body portion **108** and a support element perimeter **110**. The support element perimeter **110** defines an edge of the support element body portion **108**.

[0036] In a preferred embodiment, a plurality of apertures **112** are disposed in the support element body portion **108** such that a plurality of support members **114** are defined between adjacent apertures **112**. The plurality of apertures **112** may be of any size or shape of interest, such as, but not limited to, circles, rectangles, or the like. In another embodiment, a single aperture is disposed in the support element body portion **108** such that the plurality of support members **114** are defined along and proximate to the support element perimeter **110**.

[0037] The filter element **104** is defined by a filter element body **116** and a filter element perimeter **118**. The filter element **104** is made of a suitable filter material that is a barrier to solid particulates while allowing passage of gas through the filter element **104**. In some embodiments, the filter element **104** may be constructed from a plurality of woven fibers and/or non-woven fibers. The random arrangement of the fiber density and/or fiber arrangement of the fibers, such as but not limited to cellulose fibers, is used to control the size of the micro-pores that block passage of solid particles. Solid ambient contamination particles may include, but are not limited to, dust, pollen, mold, gluten dust, peanut dust, bacteria, and/or viruses. In an example embodiment, the filter element **104** is made of a high-efficiency particulate arresting (HEPA) grade filter material. For convenience, some embodiments of the filter element **104** may be made of readily available N95 filter materials. A N95 filter meets the U.S. National Institute for Occupational Safety and Health N95 classification of air filtration, wherein at least 95% of airborne solid particles that have a mass median aerodynamic diameter of 0.3 microns are blocked by the N95 grade filter material.

[0038] Preferably, the filter element **104** is made of a flexible or semi-flexible material to facilitate fabrication of a product package filter assembly **100**. The flexibility facilitates the sealable joining of the product package filter assembly **100** to a product container as described in greater detail hereinbelow. Some embodiments may employ a rigid, or semi-rigid filter element **104**. Any suitable filter material now known or later developed may be used in the various embodiments without departing from the spirit and scope of the present invention, and are intended to be protected by the accompanying claims.

[0039] The surface film element **106** is defined by a surface film element body **120** and a surface film element perimeter **122**. The surface film element **106** is made of a suitable film material that is preferably flexible or semi-flexible.

[0040] A plurality of micro-cuts **124** are disposed on the surface film element body **120**. Each micro-cut **124** is a cut made through the entirety of the surface film element body **120**. In a

preferred embodiment, the micro-cuts **124** may be a semicircular cut. In other embodiments, the micro-cuts **124** may be a curved cut, an oval cut, a square cut, a rectangular cut, a triangular cut, and/or another cut wherein the ends of the cut are not joined together, thereby defining a fold line **126**. The micro-cuts **124** may be of any predefined length (perimeter length). In some alternative embodiments, a micro-cut **124** may be a straight line cut of a predefined length. The micro-cuts **124** may be made in any suitable manner using any suitable cutting means, such as a die and punch, a cutting edge, a laser, or the like. The micro-cuts **124** may be made as part of the film fabrication process in some embodiments.

[0041] In the various embodiments, each micro-cut **124** and a corresponding fold line **126** cooperatively define a hinged surface portion **128**. When an internal pressure on the interior side of the hinged surface portion **128** exceeds the external air pressure on the opposing side of the hinged surface portion **128**, the hinged surface portion **128** moves outwardly towards the lower pressure exterior so that gas may pass through the micro-cut **124** from the high pressure region to the lower pressure exterior region.

[0042] As noted above, the micro-cut **124** opens in response to a pressure difference to allow gas to pass from a high pressure region to the lower pressure exterior region. In the various embodiments, when the pressure is the same on both sides of the surface film element body **120**, the micro-cuts **124** remain closed to prevent passage of solid particles through the micro-cuts **124**. An unexpected benefit provided by layering the surface film element **106** disposed over and on top of the filter element **104** is that if the exterior pressure exceeds the pressure on the interior side of the hinged surface portion **128**, the hinged surface portion **128** cannot open because the filter element **104** blocks the hinged movement of the hinged surface portion **128**. Accordingly, ambient contamination particles are prevented from passing, or at least cannot easily pass through, the closed micro-cuts **124** on the surface film element **106**.

[0043] In the example embodiment illustrated in FIG. **1**, the micro-cuts **124** are semicircular. The ends of the semicircular cut **124** are relatively close together. For example, with respect to a semicircular cut **124**, the angle α corresponds to a predefined angle selected from an angle range of between 15° to 20° , though any suitable angle may be used in the various embodiments to predefine the length of the fold line **126**. An unexpected benefit of a relatively small predefined angle α is that the length of the fold line **126** is relatively short. Accordingly, when generated gas passes through the contaminant filter assembly **100**, the relatively short fold line **126** exerts less resistance to the upward movement of the hinged surface portion **128** (as compared to the resistance from a longer fold line **126**).

[0044] During fabrication, in one embodiment, a small perimeter edge portion of a circular die is removed to define the fold line **126** in accordance with the angle α . When the die and corresponding punch are used to create the semicircular cut **124**, a relatively small amount of the film is not cut, thereby defining the fold line **126**. When a plurality of dies are aligned together on a die support structure, and when a corresponding plurality of aligned punch holes are oriented below the dies, the surface film element **106** may be placed between the dies and the corresponding punch holes such that the micro-cuts **124** are formed when the dies and punch holes are urged together. In one example process, the dies and punch holes may be disposed on roller structures so that a continuous roll or length of the surface film element **106** may be through the rolling dies and punch hole roles to continuously for a series of micro-cuts **124**.

[0045] Other cutting means may be used in alternative embodiments. For example, but not limited to, a laser system may be used to create the micro-cuts **124**. Further, the laser system may be configured to form any shape and/or dimension of interest for the micro-cuts **124**.

[0046] FIG. **2** is a perspective view of a nonlimiting example embodiment of the product package filter assembly **100** with the support element **102**, the filter element **104**, and the surface film element **106** of FIG. **1** sealably joined together during a first fabrication process. In this example embodiment, the same sized and shaped perimeters **110**, **118**, **122** of the support element **102**, the

filter element **104**, and the surface film element **106** are sealably joined together during the first fabrication process.

[0047] During a first example fabrication process, the perimeters **110**, **118**, **122** of the support element **102**, the filter element **104**, and the surface film element **106** are placed together in an ordered layer, stack, or the like. Then, a sealing means **202**, such as a bead or strip of adhesive. Is applied along the perimeters **110**, **118**, **122** to sealably join the support element **102**, the filter element **104**, and the surface film element **106** together into a unibody structure to form the product package filter assembly **100**. In the above-described first fabrication processes, the support element **102**, the filter element **104**, and the surface film element **106** are sealably joined together at the same time. In alternative embodiments, a selected two of the support element **102**, the filter element **104**, or the surface film element **106** may be sealably joined together, or at select locations, wherein the other layer is later sealably joined to finish the first fabrication process.

[0048] When the support element **102**, the filter element **104**, and the surface film element **106** are made of flexible or semi-flexible materials, the product package filter assembly **100** may be shaped as needed to be sealably joined to a product container **204**. In some embodiments if the product package filter assembly **100**, a sheet or continuous strip of the layered or stacked support element **102**, filter element **104**, and surface film element **106** is formed. Here, the lateral perimeters are sealably joined together, or at least are joined together at selected locations. Then, based on the particular characteristics of the product container that the product package filter assembly **100** is to be sealably joined to, the sheet or strip of layered elements **102**, **104**, **106** may be cut to a desired shape and/or length of interest (which typically corresponds to the dimensions of an opening in the product container that is to be sealably covered by the contaminant filter assembly **100**).

[0049] In the simplified nonlimiting hypothetical example illustrated in FIG. 2, a nonlimiting example product container **204** is conceptually illustrated as an open ended box (no top surface) wherein the upper edges **208** of the sides of the product container **204** are exposed for securing to the joined perimeters **110**, **118**, **122** of the product package filter assembly **100**. Here the size and shape of the joined perimeters **110**, **118**, **122** correspond to the size and shape of the upper edges **208** of the product container **204**.

[0050] During a second fabrication process, the product package filter assembly **100** is placed over the opening of the product container **204** such that the perimeters **110**, **118**, **122** of the product package filter assembly **100** are aligned with the upper edges **208** of the product container **204**. Once in place, the perimeters **110**, **118**, **122** may be sealably joined to the upper edges **208** of the of product container **204** using any suitable sealing means, thereby creating an airtight seal (cooperatively created by sealably joining the lower surface of the surface film element **106** to the product container **204**) such that ambient contaminants and particles are prevented from passing into the interior **206** of the product container **204**. In some applications the above-described first fabrication process and this second fabrication process may be concurrently performed together at the same time.

[0051] In some embodiments, the size of the support element **102** may be greater than the sizes of the filter element **104** and the surface film element **106**. That is, the perimeter **110** may be larger in one or more selected directions (the longitudinal axis and/or the lateral axis) such that the perimeters **118**, **122** are sealably joined to the surface of the support element **102**. An unexpected advantage of having a larger sized support element **102** is that an attachment of the support element **102**, or a portion thereof, to a product container **204** may be more easily made. In such embodiments, the filter element **104** and/or the surface film element **106** do not need to be exactly centered over the opening of the product container **204** during the second fabrication process. Further, the filter element **104** and/or the surface film element **106** may be smaller than the opening of the product container **204**, thereby reducing material costs. The area of the filter element **104** and the surface film element **106** only needs to be sufficiently large to vent the generated gases when the product residing in the product container **204** is heated.

[0052] FIG. 3 is a perspective view of the product package filter assembly **100** and the product container **204** after they have been sealably joined together after conclusion of the second fabrication process. In this example embodiment, a continuous strip or bead of adhesive has been applied to sealably join the perimeters **110**, **118**, **122** of the product package filter assembly **100** to the upper edges **208** of the product container **204**. Accordingly, a product placed within the interior **206** (FIG. 2) of the product container **204** is secured within the product container **204** in an airtight manner such that solid particles outside of the product container **204** cannot pass into the interior **206** of the product container **204**.

[0053] In some applications, the product container **204** may be optionally transparent or partially transparent so that a user can view the product residing in the interior **206** of the product container **204**. Accordingly, the manufacturer can view the product in the product container **204** for quality assurance, and/or the user may view the product during its heating or cooking. In some embodiments, the support element **102**, the filter element **104**, and/or the surface film element **106** may also be made of transparent or semi-transparent materials to facilitate the viewing of the product by the manufacturer and/or the user.

[0054] An unexpected benefit provided by the sealably joined perimeters **110**, **118**, **122** is that the filter element **104** (which is sandwiched between the support element **102** and the surface film element **106**) is securely held in place during later stages of fabrication where the product package filter assembly **100** is sealably joined with the product container **204**. During transportation of the product container **204** and the sealably joined product package filter assembly **100**, and/or use of the product container **204** when a product (not shown) residing in the interior **206** of the product container **204** is heated, the filter element **104** is securely held in place. That is, the inner surfaces of the plurality of support members **114** of the support element **102** and the surface film element body **120** frictionally secure the filter element **104** in place. In some embodiments, small amounts of adhesive or another suitable sealing means may be applied to the inner surfaces of the plurality of support members **114** at selected locations, or at selected locations of the inner surface of the surface film element body **120**, to further secure the filter element **104** in place.

[0055] Further, in embodiments where the support element body portion **108** is semi-rigid or rigid, the size and shape of the support element body portion **108** may be defined to correspond to the opening of the product container **204**, and may even serve as part of the product container **204** itself. The semi-rigid or rigid support element **102** results in a semi-rigid or rigid product package filter assembly **100** after the first fabrication process. This semi-rigidity or rigidity of the assembled product package filter assembly **100** facilitates easier joining of the product package filter assembly **100** to the product container **204** by a fabricating person or a robotic system.

[0056] In some embodiments, the surface film element **106**, or selected portions thereof, may be made of a semi-rigid or rigid material. In such embodiments, the surface film element **106** may provide enhanced integrity of the airtight seal between the product package filter assembly **100** and the product container **204**, and/or provide additional protective strength to the product package (cooperatively defined by the product container **204** and the product package filter assembly **100**). In other embodiments, additional film layers may be used without departing from the spirit of the invention and the scope of the accompanying claims.

[0057] FIG. 4 is an exploded view of a product package filter assembly **100** that further comprises a removeable moisture barrier film **402** that is an impermeable moisture barrier. In this simplified example, the removeable moisture barrier film **402** is defined by a moisture barrier perimeter **404** that approximately, or exactly, corresponds to the perimeters **110**, **118**, **122**.

[0058] Accordingly, the removeable moisture barrier film **402** is approximately the size and shape of the support element **102**, the filter element **104**, and the surface film element **106**. In other embodiments, the size of the removeable moisture barrier film **402** is at least large enough to cover the plurality of micro-cuts **124** disposed over the surface of the surface film element **106**.

Preferably, the removeable moisture barrier film **402** is made of a flexible or semi-flexible material.

[0059] The moisture barrier film **402** may be sealably joined to the top surface of the surface film element **106** using any suitable releasing sealing means, proximate to the moisture barrier perimeter **404**, prior to sealably joining the elements **102**, **104**, **106** to the product container **204** (FIG. 2). Alternatively, the moisture barrier film **402** may be sealably joined to the top surface of the surface film element **106** after the elements **102**, **104**, **106** are sealably joined to the product container **204**. In some embodiments, the moisture barrier film **402** may be sealably joined to one or more exterior surfaces of the product container **204**. In some embodiments, the moisture barrier film **402** may itself be a separate moisture and airtight package that encases the product container **204** with its sealably joined product package filter assembly **100**.

[0060] In an example embodiment, the removeable moisture barrier film **402** is releasably secured to the top surface of the surface film element **106** proximate to the bead or strip of sealing means **202** (FIG. 2) that is applied along the perimeters **110**, **118**, **122** to sealably join the support element **102**. That is, the sealing means is releasable so that the moisture barrier film **402** is removeable from the surface film element **106**. The removeable moisture barrier film **402** may be secured using another bead or strip of adhesive is applied proximate to the sealing means **202** or proximate to the perimeters **110**, **118**, **122** to sealably join the removeable moisture barrier film **402** to the top surface of the support element **102**.

[0061] Since the removeable moisture barrier film **402** has a size that is sufficiently large enough to cover all of the micro-cuts **124**, moisture from the ambient environment cannot enter into and/or leave the interior **206** of the product container **204**. Prior to use, the removeable moisture barrier film **402** is removed from the surface film element **106** so that gases generated during the heating of the product in the product container **204** may vent out of the micro-cuts **124**.

[0062] Some embodiments of the removeable moisture barrier film **402** include a pull tab **404** that is readily grasped by the user. Grasping the optional pull tab **404**, the user can then peel back the removeable moisture barrier film **402** away from the surface of the surface film element **106** prior to use. The removeable moisture barrier film **402** can then be completely, or partially, removed.

[0063] FIG. 5A is a cross sectional view of the product container **204**, the sealably joined support element **102**, filter element **104**, and surface film element **106**, and the removeable moisture barrier film **402**. A product **502** that is to be heated, thereby releasing a gas into the interior **206** of the container **204**, resides within the interior **206** of the product container **204**.

[0064] In this simplified hypothetical example, the sealing means **202**, such as an example bead or strip of adhesive, has been applied along the perimeters **110**, **118**, **122** to sealably join the support element **102**, the filter element **104**, and the surface film element **106**. The removeable moisture barrier film **402** has separately been sealably joined to the top surface of the surface film element **106** using a selected sealing means **504**. After a product **502** has been placed within the interior **206** of the product container **204**, the product package filter assembly **100** is oriented over the opening of the product container **204**, and then is sealably joined to the product container **204** using a third sealing means **506**.

[0065] FIG. 5B is a cross sectional view of an embodiment wherein the dimension of the filter element **104** and surface film element **106** are less than the dimensions of the product container **204** and the sealably joined support element **102**. The lower perimeter of the support element **102** is secured to the container **204** using the sealing means **506**. The sealing means **202** secures the perimeters of the surface film element **106**, and optionally the filter element **104**, to the upper surface of the support element **102**. A removeable moisture barrier film **402** is sealably joined to the top surface of the support element **102** using a selected sealing means **504**. In an alternative embodiment, the removeable moisture barrier film **402** is sealably joined to a selected exterior surface of the product container **204** using a selected sealing means **504**. The sealing means **504** may include, but is not limited to, an adhesive, a heat seal, a pressure seal, or any other suitable sealing means.

[0066] A product **502** resides within the product container **204**. When the product **502** is heated,

such as during cooking or sterilization, moisture in the product **502** and/or in the product container **204** vaporizes into a gas. The generated gas increases pressure within the product container **204**. When the increasing pressure within the product container **204** exceeds the ambient pressure outside of the product container **204**, the gasses vent out of the product container **204** through the hinged surface portions **128** as the hinged surface portions **128** are urged outwardly from the surface of the surface film element **106**.

[0067] As the gases vent out of the product container **204**, the filter element **104** prevents any ambient contaminants or particles from entering into the interior of the product container **204**. Accordingly, the product **502** cannot be contaminated with ambient particles.

[0068] For example, if the product **502** is a food product that is cooked while in the product container **204**, and if the user is gluten sensitive, peanut allergic, etc., then the user would be protected since such particles cannot enter into the interior **206** of the product container **204**. An unexpected advantage occurs when, for example, the food is cooked or partially cooked in a facility where other gluten and/or peanut based products are or have been prepared. The product **502** may be placed into the interior **206** of the product container **204** in a contaminant free environment. The product package filter assembly **100** is sealably joined to the product container **204** during a packaging process that is performed in the contaminant free environment, such as a gluten free and/or peanut free facility. Then, the product **502** may be transported to, and then cooked or partially cooked, in another facility that may not necessarily be contaminant free.

[0069] Another example is sterilization of medical equipment and/or substances. Here, some small amount of fluid (moisture) is introduced into the interior **206** of the product container **204** during the packaging process. For example, an optional moisture member **508** with some predefined amount of liquid may be placed into the interior of the product container **204**. The moisture in the moisture member **508** transforms into a gas when heated. An unexpected advantage is that, after packaging by the embodiments of the contaminant filter assembly **100**, the surgical equipment may be transported to a first location for sterilization. As the product container **204** and the sterilized surgical tool cool, the filter element **104** prevents entry of any contaminants or particles back into the interior **206** of the product container **204**. After sterilization, an optional removeable moisture barrier film **402** may be secured to the product container **204** and/or the surface film element **106** to cover the micro-cuts **124**. Then, the sterilized product container **204** may be transported to a second remote location for use during a surgical procedure. The product package filter assembly **100** prevents entry of any contaminants or particles back into the interior **206** of the product container **204** during transportation to the second remote location. The airtight sealed product container **204** may be stored at the second location until use (or even transported to a third remote location of use).

[0070] FIG. **6** is a perspective view of a product container **204** that has a relatively small opening **602** disposed on the surface of the product container **204**. Here, the opening **602** is conceptually illustrated as being disposed on the top surface **604** of the product container **204**. However, the opening **602** may be located on any desired surface, such as the side and/or bottom of the product container **204**. The size and shape of the opening **602** may be defined based on any characteristic of interest of the product, the product container **204**, and/or the product package filter assembly **100**. The opening **602** may be of any size or shape of interest, and only needs to be large enough to permit adequate venting of gas that is generated during the heating process.

[0071] FIG. **7** is a perspective view of a gusseted type of product container **204** provisioned with a product package filter assembly **100** sealably joined to a port or opening of a predefined size and dimension that is disposed in the gusset bag **204** at a predefined location.

[0072] FIG. **8** is a perspective view of a stand-up type of product container **204** provisioned with a product package filter assembly **100** sealably joined to a port or opening disposed in the stand-up type of bag **204**. The stand-up type of bag **204** (and/or the gusseted type of product container **204** of FIG. **7**) may optionally include a re-sealable top opening **802**. One skilled in the art appreciates

that embodiments of a product package filter assembly **100** can be sealably joined to any type of product container **204**. All such embodiments are intended to be disclosed herein and to be protected by the accompanying claims.

[0073] One skilled in the arts appreciates that as a packaged product cools, the volume of any gas within the product container **204** will decrease as the temperature decreases. In the absence of the contaminant filter assembly **100**, a sealed package may collapse on itself during cooling since a pressure difference between the interior of the package and the ambient environment results in a vacuum or partial vacuum in the package. A collapse of the package may be undesirable, such as if the collapsing package comes into contact with the product. The collapsing package might deform the product. The collapsing package may also negatively impact the visual appearance of the package. However, embodiments of the contaminant filter assembly **100** permit gas and/or moisture to enter into the product container **204** during cooling. An unexpected advantage provided by embodiments of the contaminant filter assembly **100** is the prevention, or at least partial prevention, of a collapse of the product container **204** during cooling as pressure equalizes.

[0074] FIG. **9A** is a perspective view of a product package filter assembly **100** wherein a length of the surface film element **106** (along the longitudinal axis) is greater than the length of the filter element **104**. In this embodiment, the optional support element **102** is omitted. The center portion **106a** of the surface film element **106** is defined by a plurality of hinged surface portions **128**. A bead or strip of adhesive **202** is applied between a lower surface of the surface film element **106** and the filter element **104** to sealably join the filter element **104** and to the lower surface of the center portion **106a** of the surface film element **106**. In this example embodiment, the surface film element **106** is flexible or semi-flexible. Other sealing means may be used in alternative embodiments.

[0075] A first side portion **106b** extends outwardly from the sealably joined filter element **104** by a predefined distance **D1**. A second side portion **106c** extends outwardly in an opposing direction from the sealably joined filter element **104** by a predefined distance **D2**. The distances **D1**, **D2** may be the same or may be different, depending upon the packing application of interest.

[0076] FIG. **9B** is a cross sectional view of the product package filter assembly **100** of FIG. **9A** secured to a tray type product container **204**. In this example embodiment, the perimeter portion **122c** is sealably joined to a first edge **208** of the tray **204** using a bead or strip of adhesive **506**. The opposing perimeter portion **122b** is also sealably joined to an opposing second edge **208** of the tray **204**. One skilled in the art appreciates that other portions of the product package filter assembly **100** are secured to the tray **204** to form an airtight seal. Other sealing means may be used in alternative embodiments.

[0077] As the product **502** is heated, the gas **902** is generated. The gas **902** is vented out through the hinged surface portions **128**. The hinged surface portions **128** are urged outwardly by the venting gas **902**. When the generation of the gas **902** ends, then the hinged surface portions **128** may relax or partially relax, thereby falling onto the top of the filter element **104**. Even if an airtight seal is not created by the relaxed hinged surface portions **128**, contaminants and/or particles cannot enter into the interior **206** of the product container **204** because of the filtering effect of the filter element **104**.

[0078] FIG. **10A** is a perspective view of a product package filter assembly **100** wherein a width of one side of the support element **102** (along the lateral axis) is greater than the corresponding width of the filter element **104** and the surface film element **106**. The first side portion **102a** of the support element **102** is defined by a plurality of apertures **112** (not visible). The size of the first side portion **102a** is larger than a size of the filter element **104** and the surface film element **106**. Also, a width of the second side portion **102b** is longer than the corresponding width of the first side portion **102a** by a predefined distance **Da**. A bead or strip of adhesive **202** is applied between a lower surface of the surface film element **106** and an upper surface of the filter element **104** to sealably join the filter element **104** and to the lower surface of the center portion **106a** of the

surface film element **106** along the perimeter edges **110a** and **110b**. The length of the support element **102**, in this non-limiting example embodiment, is greater than the corresponding length of the filter element **104** and the surface film element **106** by a predefined distance **Db**. In this example embodiment, the surface film element **106** is flexible or semi-flexible. Other sealing means may be used in alternative embodiments.

[0079] FIG. **10B** is a perspective view of the product package filter assembly **100** of FIG. **10A**. Here, the first side portion **102a** and the second side portion **102b** are folded over around a fold line **1002**. The perimeter edges **110a** and **110b** are sealably joined together using a selected sealing means. Some embodiments may employ a bead or strip of adhesive, a crimp, a heat seal, or other sealing means **1004** to form a pouch-like container that retains the product **502**. That is, the support element **102** is folded over and sealably joined to create the equivalent product container **204**.

[0080] In some applications, the support element **102** may be folded around the product **502** and then the edges **110a**, **110b** may be sealably joined. Then, the ends **110c** and **110d** may each be sealably joined to secure the product **502** in an air tight container. Any suitable sealing means may be used in the various embodiments.

[0081] Alternatively, the support element **102** may be initially folded to sealably join the edges **110a**, **110b**, thus forming an empty pouch-like container **204**. The product **502** may then be inserted into the interior of the pouch-like container. Then, the ends **110c** and **110d** may each be sealably joined to secure the product **502** in an airtight container. In some applications, a continuous length of the empty pouch-like container may be formed, and then later cut as needed to a desired length.

[0082] FIG. **11** is an exploded view of an alternative embodiment of the product package filter assembly **1100** before fabrication. Embodiments of the product package filter assembly **1100** comprise an optional support element **102**, a filter element **104**, and a surface film element **106**. The surface film element **106** is defined by a surface film element body **120** and a surface film element perimeter **122**. The surface film element **106** is made of a suitable film material that is preferably flexible or semi-flexible.

[0083] A plurality of cuts **1124** are disposed on the surface film element body **120**. Each cut **1124** is a cut made through the entirety of the surface film element body **120** to form an aperture **1126**. In a preferred embodiment, the cuts **1124** may be a circular cut. In other embodiments, the cuts **1124** may be a curved cut, an oval cut, a square cut, a rectangular cut, a triangular cut, and/or another cut wherein the aperture **1126** is defined. The cuts **1124** may have any predefined length (perimeter length). The cuts **1124** may be made in any suitable manner using any suitable cutting means, such as a die and punch, a cutting edge, a laser, or the like. The cuts **1124** may be made as part of the film fabrication process in some embodiments.

[0084] In an example embodiment of the product package filter assembly **1100**, the support element **102**, the filter element **104**, and the surface film element **106** are sealably joined together using any suitable process. Preferably, a removeable impermeable moisture barrier film **402** (FIG. **4**) is sealably joined to the top surface of the surface film element **106** using any suitable releasing sealing means. The removeable impermeable moisture barrier film **402** ensures that contaminants cannot enter past the product package filter assembly **1100** prior to use. In some embodiments, the moisture barrier film **402** may itself be a separate moisture and airtight package that encases the product container **204** with its sealably joined product package filter assembly **100**.

[0085] In some embodiments, the impermeable moisture barrier film **402** sealing means seals the entire perimeter of the impermeable moisture barrier film **402** to the top surface of the product package filter assembly **1100**. In another embodiment, at least a part of the sealing means releasably secures the removeable impermeable moisture barrier film **402** to the product package filter assembly **1100**. Here, a portion of the removeable impermeable moisture barrier film **402** may be released (and peeled backwardly) so that the package assembly may be used.

[0086] In this example embodiment, the removeable moisture barrier film **402** is releasably secured to the top surface of the surface film element **106**. Prior to use, the removeable moisture barrier

film **402** may be removed from, or at least partially pulled away from, the top surface of the surface film element **106**. During heating of a product residing in the product container having the product package filter assembly **1100**, gasses generated in the product container vent out through the product package filter assembly **1100**, via the apertures **1126**, while the filter element **104** prevents entry of any ambient contamination particles into the product container.

[0087] One skilled in the art appreciates that the various embodiments of the product package filter assembly comply with the well-known protocol, Hazard Analysis Critical Control Points (HACCP). HACCP provides the system for monitoring the total food processing, from harvesting to consumption. The purpose of HACCP is to reduce the risk of foodborne illness by identifying and controlling potential problems before they occur. Contamination particles or contaminants, as defined herein, are unacceptable physical particles, debris, protein particles, organic dusts (dried sewage or fecal dust), microbial growth (such as yeasts, molds, fungi and parasites), viral contaminants, bacterial contaminants, persistence of toxins and/or survival of microorganisms that jeopardize food safety.

[0088] The various embodiments of the product package filter assembly described herein facilitate prevention of contamination to packaged food items and/or other packaged items, such as medical equipment, sensitive electronics, or the like. When a HACCP is in place to monitor food product quality, embodiments of the product package filter assembly facilitate compliance with HACCP protocols.

[0089] One skilled in the art appreciates that alternative embodiments of the product package filter assembly **1100** may be made and/or used in the manner as described hereinabove for the product package filter assembly **100**. All such variations are intended to be disclosed herein and to be protected by the accompanying claims.

[0090] While the preferred embodiment of the invention has been illustrated and described, as noted above, many changes can be made without departing from the spirit and scope of the invention. Each disclosure of a component preferably having a feature or characteristic is intended to also disclose the component as being devoid of that feature or characteristic, unless the principles of the invention clearly dictate otherwise. Accordingly, the scope of the invention is not limited by the disclosure of the preferred embodiment. Instead, the invention should be determined entirely by reference to the claims that follow. It should also be noted that the claim dependencies or combinations of elements recited in the claims does not reflect an intention to forgo claiming other subject matter disclosed herein. Instead, this disclosure is intended to also disclose the subject matter of any combination of any two or more of the claims, such that subsequent claim sets may recite that any one of the dependent claims depends from any other one or more claims, up to and including all other claims in the alternative (such as “The apparatus or method of any one of the preceding or subsequent claims . . .”). This disclosure is also intended to disclose the subject matter of any one of the dependent claims, as if it was an independent claim, with or without all or a portion of the subject matter of the original independent claim(s) or any other subject matter disclosed herein.

Claims

1. A product package filter assembly, comprising: a support element, wherein the support element is defined by: a support element body; a support element perimeter; and at least one support element aperture disposed in the support element body; a filter element, wherein the filter element is defined by: a filter element body; and a filter element perimeter; a surface film element, wherein the surface film element is defined by: a surface film element body; at least one cut disposed in the surface film element body that extends through the surface film element body to define at least one surface film element aperture; and a surface film element perimeter; and a sealing means, wherein the support element, the filter element, and the surface film element are arranged in a layer with the

filter element disposed between the support element and the surface film element, and wherein at least the support element perimeter and the surface film element perimeter are sealably joined together using the sealing means to form an airtight barrier between the support element perimeter and the surface film element perimeter.

2. The product package filter assembly of claim 1, wherein the at least one cut is one of a plurality of cuts disposed in the surface film element body to define a plurality of surface film element apertures.

3. The product package filter assembly of claim 2, wherein in response to heating a product residing in an interior of a product container that is sealably joined to the product package filter assembly, a gas is generated that increases an interior pressure within the interior of the product container so that the interior pressure becomes greater than an ambient pressure, wherein the generated gas vents out through the plurality of surface film element apertures of the product package filter assembly, and wherein the filter element prevents ambient particles from entering into the interior of the product package.

4. The product package filter assembly of claim 3, wherein after completion of the heating of the product, the interior of the product package cools such that the interior pressure becomes less than an ambient pressure, and wherein the filter element prevents ambient particles from entering into the interior of the product package as ambient air enters into the interior of the product package as the product package cools.

5. The product package filter assembly of claim 1, wherein the support element perimeter, the filter element perimeter, and the surface film element perimeter are sealably joined together using the sealing means.

6. The product package filter assembly of claim 5, wherein in response to heating a product residing in an interior of a product container that is sealably joined to the product package filter assembly, a gas is generated that increases an interior pressure within the interior of the product container so that the interior pressure becomes greater than an ambient pressure, wherein the gas vents out of the product container into an ambient region around the product container firstly via the at least one support element aperture disposed in the support element body, secondly via the filter element body, and thirdly via the at least one surface film element aperture disposed in the surface film element body.

7. The product package filter assembly of claim 1, wherein the sealing means is a first sealing means, wherein the airtight barrier is a first airtight barrier, and further comprising: a second sealing means, wherein the second sealing means sealably joins a bottom surface of the support element to a surface of a product container so that a second airtight barrier is formed between at least the perimeter of the support element and the surface of the product container.

8. The product package filter assembly of claim 7, wherein the at least one cut is one of a plurality of cuts disposed in the surface film element body to define a plurality of surface film element apertures, and further comprising: a moisture barrier film that is an impermeable moisture barrier and that is defined by a moisture barrier perimeter; and a third sealing means that sealably joins a top surface of the surface film element with a bottom surface of the moisture barrier proximate to the moisture barrier perimeter, wherein the third sealing means is releasable so that the moisture barrier film is removeable from the surface film element, and wherein a size of the moisture barrier film is large enough to cover the plurality of surface film element apertures disposed in the surface film element.

9. The product package filter assembly of claim 1, further comprising: a moisture member that retains a predefined amount of liquid, wherein the moisture member resides in an interior of the product container, and wherein the liquid in the moisture member transforms into a gas when heated.

10. The product package filter assembly of claim 1, wherein the sealing means is a first sealing means, wherein the airtight barrier is a first airtight barrier, and wherein the at least one micro cut is

one of a plurality of micro cuts disposed in the surface film element body, and further comprising: a moisture barrier film that is an impermeable moisture barrier and that is defined by a moisture barrier perimeter.

11. The product package filter assembly of claim 10, further comprising: a second sealing means that sealably joins a top surface of the surface film element with a bottom surface of the moisture barrier proximate to the moisture barrier perimeter, wherein the second sealing means is releasable so that the moisture barrier film is at least partially removeable from the surface film element, and wherein a size of the moisture barrier film is large enough to cover the at least one surface film element aperture disposed in the surface film element.

12. The product package filter assembly of claim 11, further comprising: a third sealing means, wherein the third sealing means sealably joins a bottom surface of the support element to a surface of a product container so that a second airtight barrier is formed between at least the perimeter of the support element and the surface of the product container.

13. The product package filter assembly of claim 1, wherein the sealing means is a first sealing means, wherein the support element is defined by a first side portion and a second side portion, wherein the at least one support element aperture is disposed in the first side portion, and wherein a fold line separates the first side portion and the second side portion, and further comprising: a second sealing means, wherein the second sealing means sealably joins a first perimeter edge of the first side portion with a second perimeter edge of the second side portion, wherein the first side portion and the second side portion cooperatively define a pouch-like container.

14. The product package filter assembly of claim 13, further comprising: a third sealing means, wherein the third sealing means sealably joins a first end of the product package filter assembly; and a fourth sealing means, wherein the fourth sealing means sealably joins a second end of the product package filter assembly, and wherein the second end of the product package filter assembly is opposing the first end of the product package filter assembly.

15. The product package filter assembly of claim 14, wherein after the third sealing means sealably joins the first end of the product package filter assembly and before the fourth sealing means sealably joins the second end of the product package filter assembly, a product is inserted into an interior of the pouch-like container.

16. The product package filter assembly of claim 1, wherein the support element aperture is one of a plurality of support element apertures disposed in the support element, wherein between each adjacent support element aperture is a support member, and wherein the support member and a corresponding lower surface of the surface film element frictionally retain the filter element.

17. A product package filter assembly, comprising: a surface film element, wherein the surface film element is defined by: a surface film element body; at least one cut disposed in the surface film element body that extends through the surface film element body to define a support element aperture; and a surface film element perimeter; a filter element, wherein the filter element is defined by: a filter element body; and a filter element perimeter; a sealing means, wherein the surface film element is disposed on top of the filter element, wherein the surface film element perimeter is sealably joined to a bottom side of the surface film element using the sealing means to form an airtight barrier between the surface film element perimeter and the bottom side of the surface film element.

18. The product package filter assembly of claim 17, wherein the sealing means is a first sealing means, wherein the airtight barrier is a first airtight barrier, and further comprising: a second sealing means, wherein the second sealing means sealably joins the bottom surface of the surface film element to a surface of a product container so that a second airtight barrier is formed between at least the perimeter of the support element and the surface of the product container.

19. The product package filter assembly of claim 18, wherein the second sealing means is proximate to the surface film element perimeter.

20. The product package filter assembly of claim 17, wherein the sealing means is a first sealing

means, wherein the airtight barrier is a first airtight barrier, and wherein the at least one micro cut is one of a plurality of micro cuts disposed in the surface film element body, and further comprising: a moisture barrier film that is an impermeable moisture barrier and that is defined by a moisture barrier perimeter; and a second sealing means that sealably joins a top surface of the surface film element with a bottom surface of the moisture barrier proximate to the moisture barrier perimeter, wherein the second sealing means is at least partially releasable so that the moisture barrier film is at least partially removeable from the surface film element, and wherein a size of the moisture barrier film is large enough to cover the at least one surface film element aperture disposed in the surface film element.
